

HOSPITAL TRIAGE AI

Deterministic Guardrails & Clinical Governance for AI-Driven Hospital Intake

- Track: Trust, Safety & Governance (Exasol AI Building Challenge)
- In-Memory Database: Exasol (Live Bed Inventory & Audit Logs)
- Clinical Brain: Google Gemini 3.5 Flash | Frontend: Streamlit

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The Core Dilemma: LLM Agility vs. Clinical Safety

Clinical Failure Modes of Pure LLMs

- **Uncontrolled Hallucination Risk:**

Pure LLMs can misclassify high-acuity chest pain or stroke symptoms, routing critical cases to general wards.

- **Cloud Dependency & System Halts:**

If external AI APIs experience network timeouts during an intake rush, emergency rooms cannot freeze operations.

- **Data Corruption & Race Conditions:**

High-concurrency admissions lead to 'ghost patients' and double-booked beds without strict database locks.

The Governed AI Architecture

- **Dual-Layer Triage Engine:**

Gemini 3.5 Flash provides natural language symptom analysis, wrapped in deterministic SQL and rule-based barriers.

- **Exasol Live Inventory Verification:**

Physical bed availability is verified in-memory BEFORE any patient record is committed to the EHR.

- **Deterministic Failsafe Mode:**

Zero single point of failure (SPOF). The app automatically falls back to clinical rules if cloud services drop.

Experiment 1: Fast-Track Emergency vs. Standard Intake Protocol

⚡ Cognitive Load Reduction in Emergencies

- **Adaptive Intake Forms:**

When set to 'Emergency (Immediate)', the UI eliminates non-critical form fields, capturing only estimated age and gender for instant triage.

- **Automated Clinical Bypass:**

Chronic conditions are automatically flagged as 'Unknown (Emergency Bypass)' so critical minutes are not lost typing administrative data.

- **Standard Mode Comparison:**

Non-emergency intakes enforce full chronic condition and demographic input for comprehensive record-keeping.

📺 Live UI Verification

The image displays two screenshots of a web application interface, likely a patient intake system, comparing two different protocols: 'Emergency (Immediate)' and 'Standard Intake'.

Top Screenshot (Emergency Mode):

- Header:** "Patient Admission & Triage"
- Triage Category:** "Emergency (Immediate)" is selected (indicated by a red dot), while "Non-Emergency (Standard Intake)" is unselected (indicated by a green dot).
- Select Patient Type:** "New Patient (Walk-in)" is selected (indicated by a red dot), while "Existing Patient (EMR Lookup)" is unselected (indicated by a green dot).
- New Patient Registration:** A yellow banner indicates "FAST-TRACK INTAKE ACTIVATED: Capturing vital details only."
- Form Fields:** The form is simplified, showing only "Full Name" (with the value "Alexander"), "Estimated Age" (with the value "30"), and "Gender" (with the value "Female").

Bottom Screenshot (Standard Mode):

- Header:** "New Patient Registration"
- Form Fields:** The form is more complex, showing "Full Name" (with the value "Alexander"), "Estimated Age" (with the value "30"), "Gender" (with the value "Female"), and a "Symptoms" field (with the value "Severe chest pain and shortness of breath").
- Footer:** A yellow banner indicates "FAST-TRACK INTAKE ACTIVATED: Capturing vital details only."

```
exapump> SELECT * FROM STARTER_KIT.PATIENT_HISTORY WHERE "Gender" = 'Female' AND "Chronic_Conditions" = 'Unknown (Emergency Bypass)';
```

Patient_ID	Name	Age	Gender	Chronic_Conditions	Recent_Surgeries
PT-15000	Alexandra	30	Female	Unknown (Emergency Bypass)	None

```
1 row
```

Experiment 2: Double-Admission Prevention & SQL Governance

🔒 Active Bed Occupancy Locking

- **The EHR Duplicate Risk:**

Human error or duplicate triage requests can assign an already admitted inpatient to a secondary bed, distorting live ward capacity.

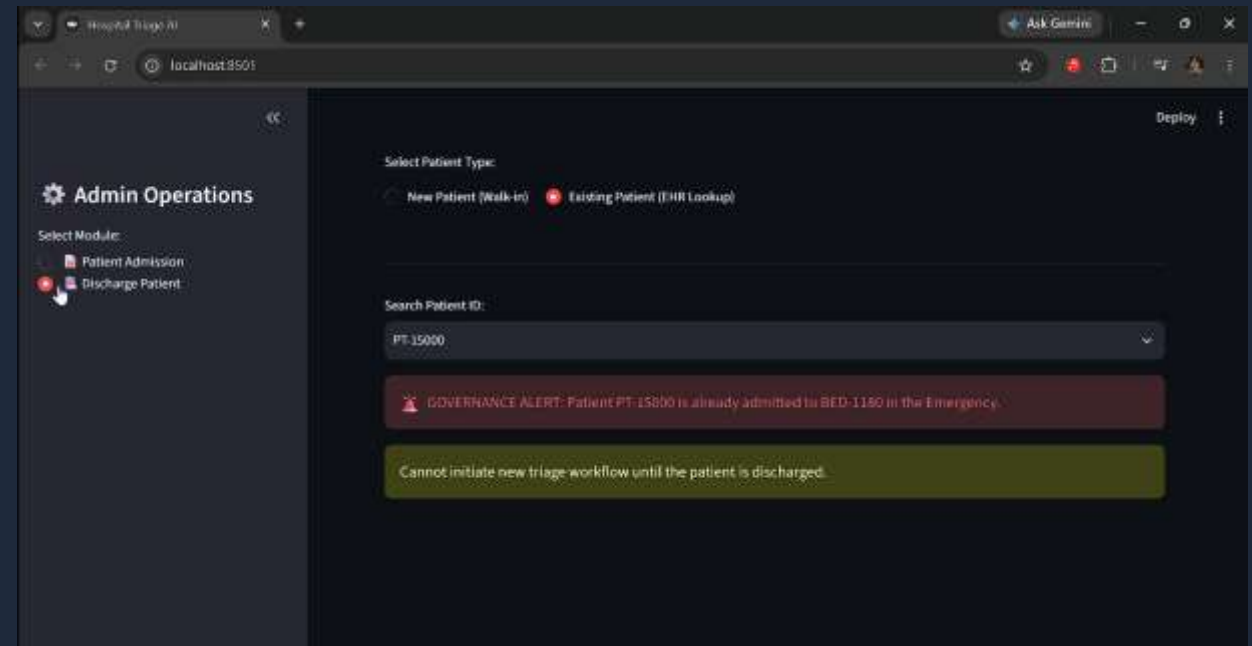
- **Pre-Triage Database Lock:**

Before loading AI chat models, the application queries Exasol: `SELECT * FROM BEDS WHERE Patient_ID = active_id AND Status = 'Occupied'`.

- **Deterministic UI Halt:**

If an active occupancy record is detected, `st.stop()` instantly terminates intake execution and raises a high-visibility Governance Alert.

📷 Live UI Verification



Experiment 3: Dynamic Surgical EHR Tracking & Real-Time Discharge

Complete Patient Lifecycle Management

- **Real-Time EHR Synchronization:**

Discharging a patient doesn't just release a bed; it captures ongoing clinical outcomes directly back into the EHR.

- **Dynamic Surgical Validation:**

If 'Was this admission for surgery?' is checked, the custom surgery name is committed to PATIENT_HISTORY. If unchecked, it cleanly defaults to 'No Prior Surgeries'.

- **Two-Way Transaction Integrity:**

Exasol atomically executes both the surgical history UPDATE and frees the bed inventory with an updated timestamp in a single atomic flow.

Live UI Verification



Admin Operations

Select Module:

- Patient Admission
- Discharge Patient

Patient Details

Room: BED-1101

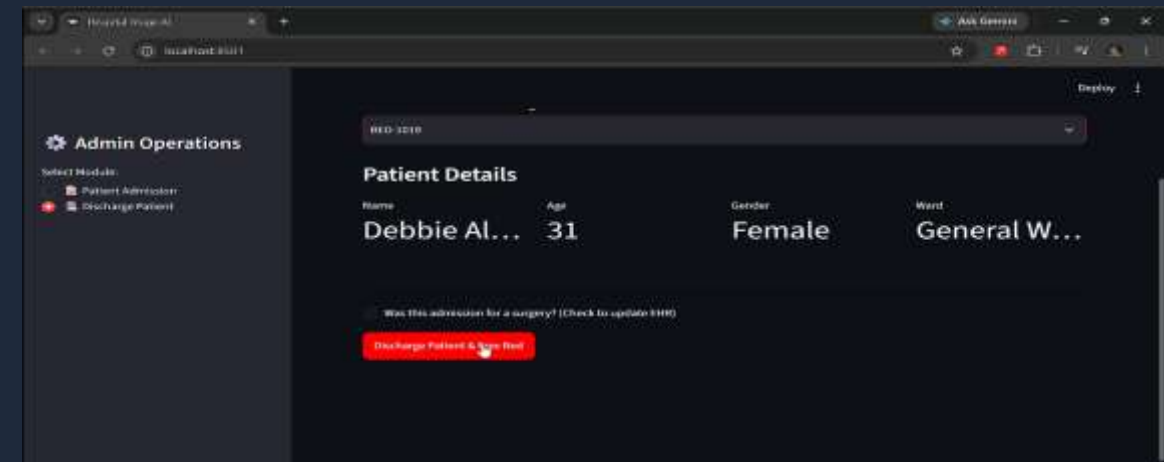
Name: Deborah ... Age: 88 Gender: Female Ward: Emergency

☒ Was this admission for a surgery? (Check to update EHR)

Enter specific surgery details:

Cardiac

Discharge Patient & Free Bed



Admin Operations

Select Module:

- Patient Admission
- Discharge Patient

Patient Details

Room: BED-1010

Name: Debbie Al... Age: 31 Gender: Female Ward: General W...

☐ Was this admission for a surgery? (Check to update EHR)

Discharge Patient & Free Bed

```
exapump> SELECT * FROM STARTER_KIT.PATIENT_HISTORY WHERE "Age" = 88 AND "Gender" = 'Female' AND "Recent_Surgeries" = 'Cardiac';
```

Patient_ID	Name	Age	Gender	Chronic_Conditions	Recent_Surgeries
PT-5978	Deborah Wright	88	Female	No Chronic Conditions	Cardiac

```
1 row
```

Experiment 4: Ward Capacity Overflows & Ghost Patient Prevention

🚫 Zero Ghost Patient Records

- **Pre-Registration Availability Checks:**

Standard systems register patients before checking bed availability. Our architecture verifies physical bed inventory in Exasol first.

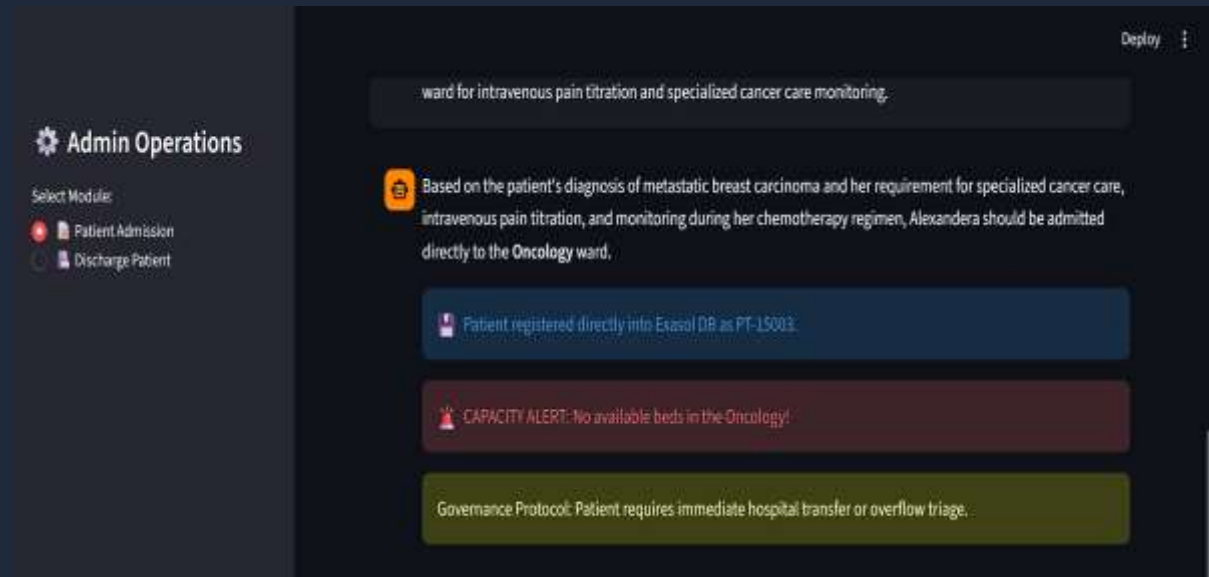
- **Automated Overflow Protocol:**

If a specialized ward (e.g., Oncology) reaches 100% capacity, the system triggers a Red Capacity Alert and activates transfer protocols.

- **Preventing Phantom Records:**

If no beds exist, the patient registration SQL INSERT is canceled completely, preventing orphan records from cluttering the hospital database.

📷 Live UI Verification



Technical Hardening: Deterministic Fallbacks & Audit Trails

⚙️ Zero-Downtime Deterministic Fallback

- **No Dead-End Error Messages:**

Passive 'try again later' messages are unacceptable in critical triage. Exceptions engage automatic deterministic rules.

- **High-Acuity Auto-Routing:**

If the Gemini API is unreachable, Emergency intakes immediately route to the Emergency Ward automatically.

- **Non-Emergency Safe Mode:**

Non-emergency cases safely default to the General Ward for standard clinical assessment without stalling intake.

🕒 Immutable Exasol Audit Logs

- **Realistic Historical Staggering:**

Synthetic_Data.py generates historical activity across a 7-day randomized window to simulate real bed usage.

- **Real-Time Execution Stamping:**

Every live transaction in Exasol_app.py stamps the exact `datetime.now()` into the Admission_Time column.

- **Full Regulatory Traceability:**

Hospital compliance auditors can trace the exact second every bed was occupied or released.

Summary & Future Governance Roadmap

Key Impact

- 100% Zero Single Point of Failure.
- Zero phantom patient database entries.
- Strict prevention of duplicate admissions.
- Instantaneous in-memory bed allocation powered by Exasol.

Roadmap

- UUID Concurrency Locks: Support for concurrent intake requests across 100+ simultaneous triage terminals.
- Poly-symptomatic Rule Engine: Expanding fallback logic for complex multi-diagnosis admissions.

Project Repo

Hospital Triage AI

GitHub Repository:
github.com/C-Priyan/hospital-triage-ai

Thank you for reviewing our submission!